

$$\begin{aligned}
 &= z'x'(y+y') + z'x(y+y') \quad (\text{Postulate 4(a)}) \\
 &= z'x'(y+y') + z'x(y+y') \quad (\text{Postulate 3(a)}) \\
 &= (z'x') \cdot 1 + (z'x) \cdot 1 \quad (\text{Postulate 5(a)}) \\
 &= z'x' + z'x \quad (\text{Postulate 2(b)}) \\
 &= z' \cdot (x' + x) \quad (\text{Postulate 4(a)}) \\
 &= z' \cdot 1 \quad (\text{Postulate 3(a)}) \\
 &= z' \cdot 1 \quad (\text{Postulate 5(a)}) \\
 &= z' \quad (\text{Postulate 2(b)})
 \end{aligned}$$

Example 3-3: Simplify the Boolean function:

Example 3.3.

Grouping

$$F = A'C + A'B + AB'C + BC$$

		BC			
		00	01	11	10
A	0		1	1	1
	1		1	1	

SA 1

Figure 3.6: Map for Example 3-3;

- Some of the terms in the function have less than three literals and are represented in the map by more than one square.
 - For example, to find the squares corresponding to $A'C$, we form the coincidence of A' (first row) and C (two middle columns) and obtain squares 001 and 011.
 - We similarly mark the squares corresponding to $A'B$, BC and $AB'C$.
 - The four squares in the center gives the literal C after simplification.
 - The remaining single square marked with a 1 in 010 is combined with an adjacent square that has already been used once. This is permissible and even desirable since the combination of the two squares gives the term $A'B$ while the single minterm represented by the square gives the three-variable term $A'BC'$.
- The simplified function is: $F = C + A'B$.

Example 3-4: Simplify the Boolean function:

$$F(x, y, z) = \sum(0, 2, 4, 5, 6)$$